Supply Chain Analysis And Prediction

Correlation Heatmap of Numeric Features

This correlation heatmap visualizes the relationships between numerical variables in the supply chain dataset.

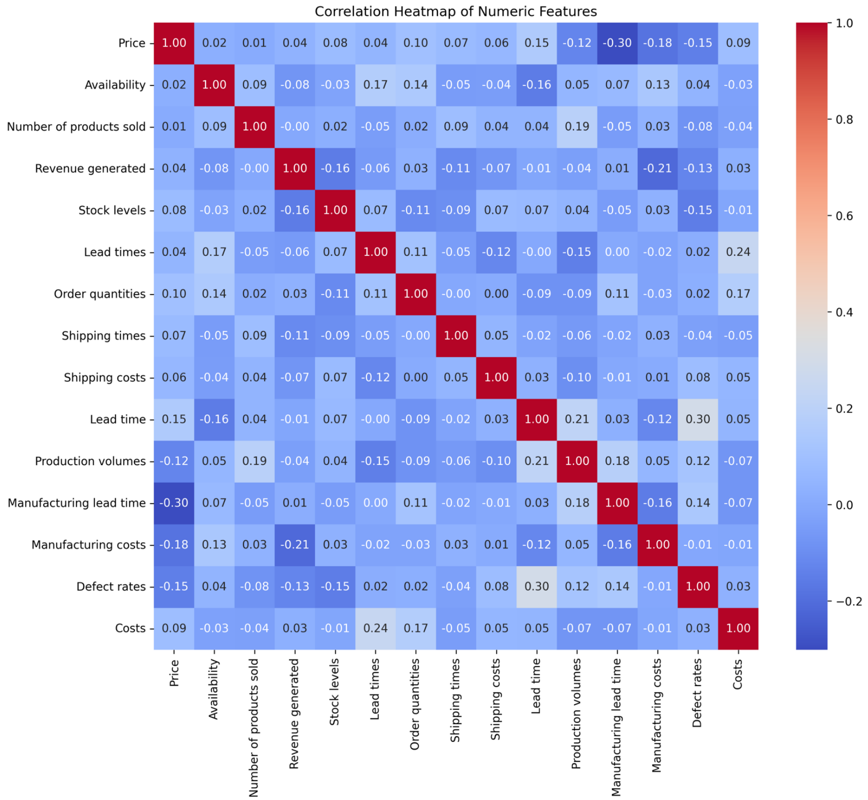
Correlation values range from **-1 to +1**, where:

+1 = Strong positive correlation

0 = No correlation

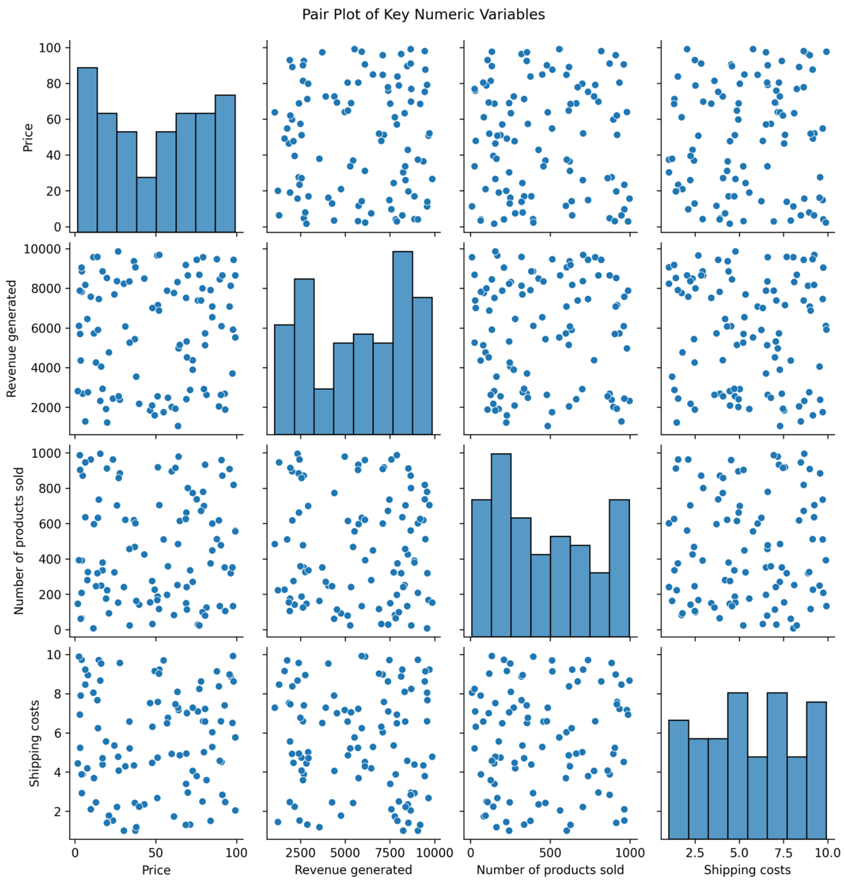
-1 = Strong negative correlation

The heatmap helps identify variables that move together and provides insights for feature selection and business decision-making.

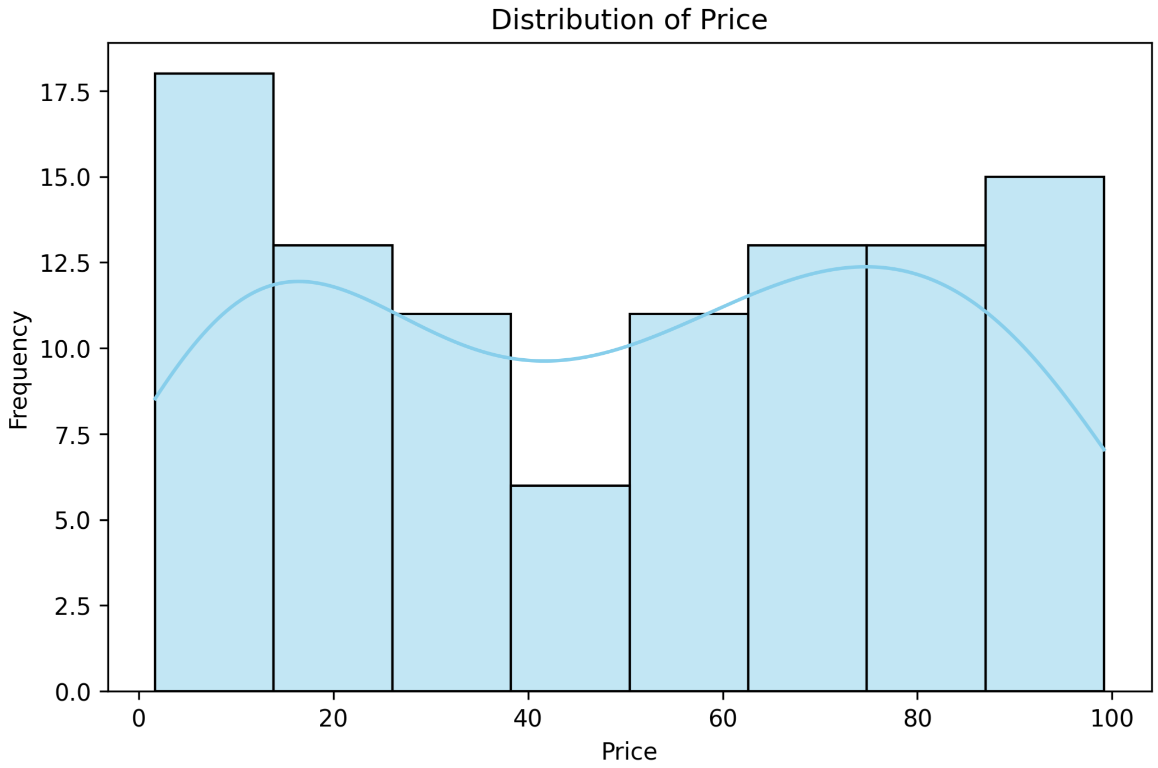


Pair Plot of Key Numeric Variables

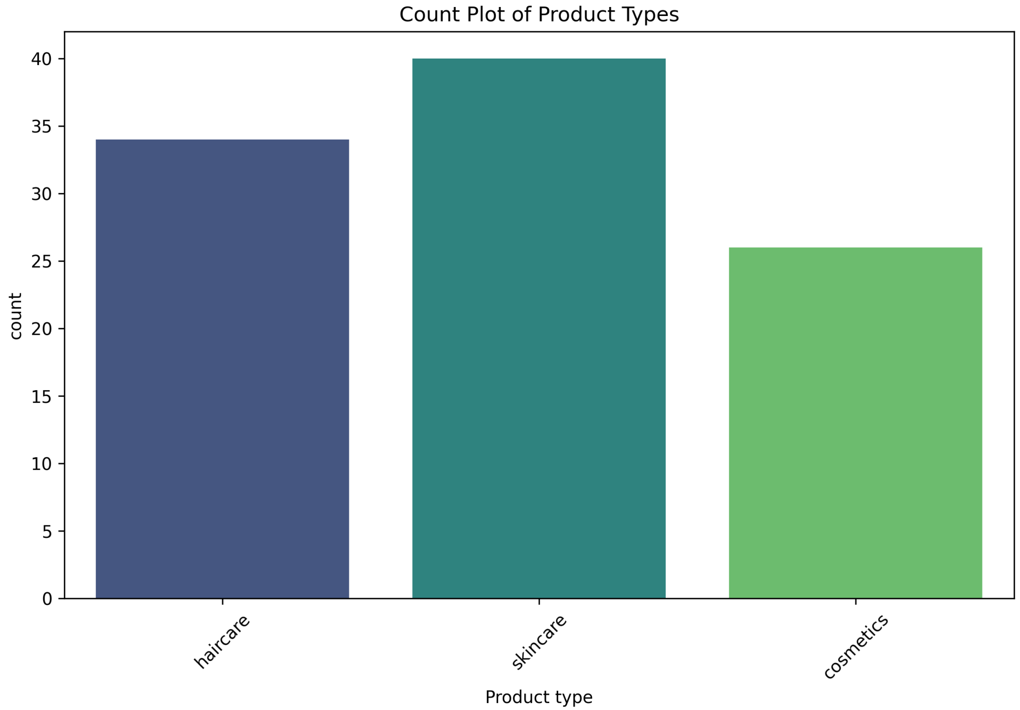
The pair plot visualizes the relationships among the key numerical variables: **Price, Revenue Generated, Number of Products Sold, and Shipping Costs**. It helps identify patterns, correlations, trends, and potential outliers within the dataset.



Histogram Analysis (Price Distribution)
Distribution of Price
This histogram visualizes the distribution of the **Price** variable in the dataset. The chart helps understand how prices are spread across different ranges and whether the data is skewed, normally distributed, or contains outliers.



Count Plot Analysis (Product Type)
Count Plot of Product Types
This count plot illustrates the distribution of products across different product categories in the dataset, including **Hair Care, Skin Care, and Cosmetics**.



Box Plot Analysis (Shipping Costs)

Box Plot of Shipping Costs

This box plot illustrates the distribution of **Shipping Costs** and helps identify the spread, central tendency, variability, and potential outliers within the dataset.

Actual vs Predicted Revenue Analysis Template

Actual vs Predicted Revenue

This scatter plot compares the **actual revenue values** with the **predicted revenue values** generated by the machine learning model. The dashed diagonal line represents the **ideal prediction line (y = x)**, where predicted values perfectly match actual values.